

SOFTWARE ENGINEERING LAB 2

Jira Hands-on – Agile Backlog Creation & Sprint Simulation

Name	DHANUSH K S
SRN	PES1UG24CS148
Section	C
Project	EV Charging Station Reservation System
Jira Board	ECSRS board
Problem Statement	#21

This report documents the Jira Agile backlog, Epics and User Stories, priorities and Story Points, two simulated one-week sprints, Burndown Charts, Burndown event details, and the required reflection.

1. Lab 2 Completion Summary

The Lab 1 requirements for the EV Charging Station Reservation System were converted into Jira Epics and User Stories, prioritized, estimated using Fibonacci Story Points, and executed through two simulated one-week sprints.

Item	Completed Evidence
Epics	4 Epics created
User Stories	8 User Stories created
Story Points	30 total points across the backlog
Sprint 1	4 stories, 14 points, all completed
Sprint 2	4 stories, 16 points, all completed
Burndown	Sprint 1: 14 → 0; Sprint 2: 16 → 0
Reflection	All 4 required questions answered

The handout requires evidence of the Jira backlog, Story Point assignments, Active Sprint view, Burndown Chart, and a brief reflection. All required sections are included below.

2. Jira Backlog – Epics and User Stories

The backlog is organized into four functional Epics and eight User Stories. The stories use the required Agile format and are estimated using Fibonacci Story Points.

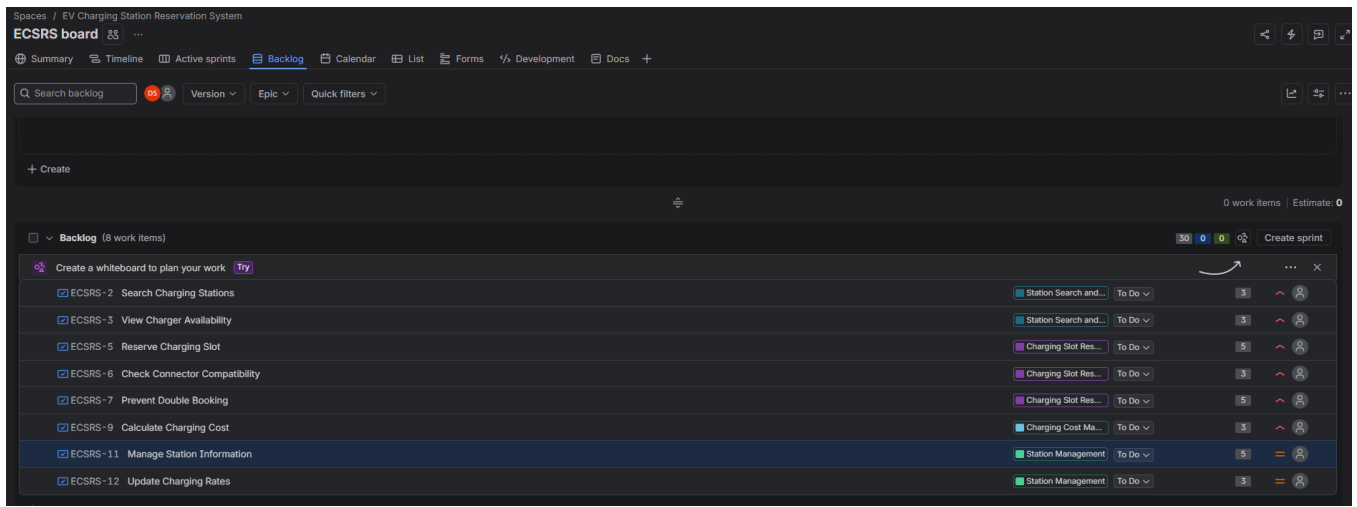


Figure 1. Jira backlog showing Epics, User Stories, priorities and Story Points.

3. Jira Sprint 1 – Active Sprint

Sprint 1 contained four stories totaling 14 story points. The stories were simulated through the To Do → In Progress → Done workflow.

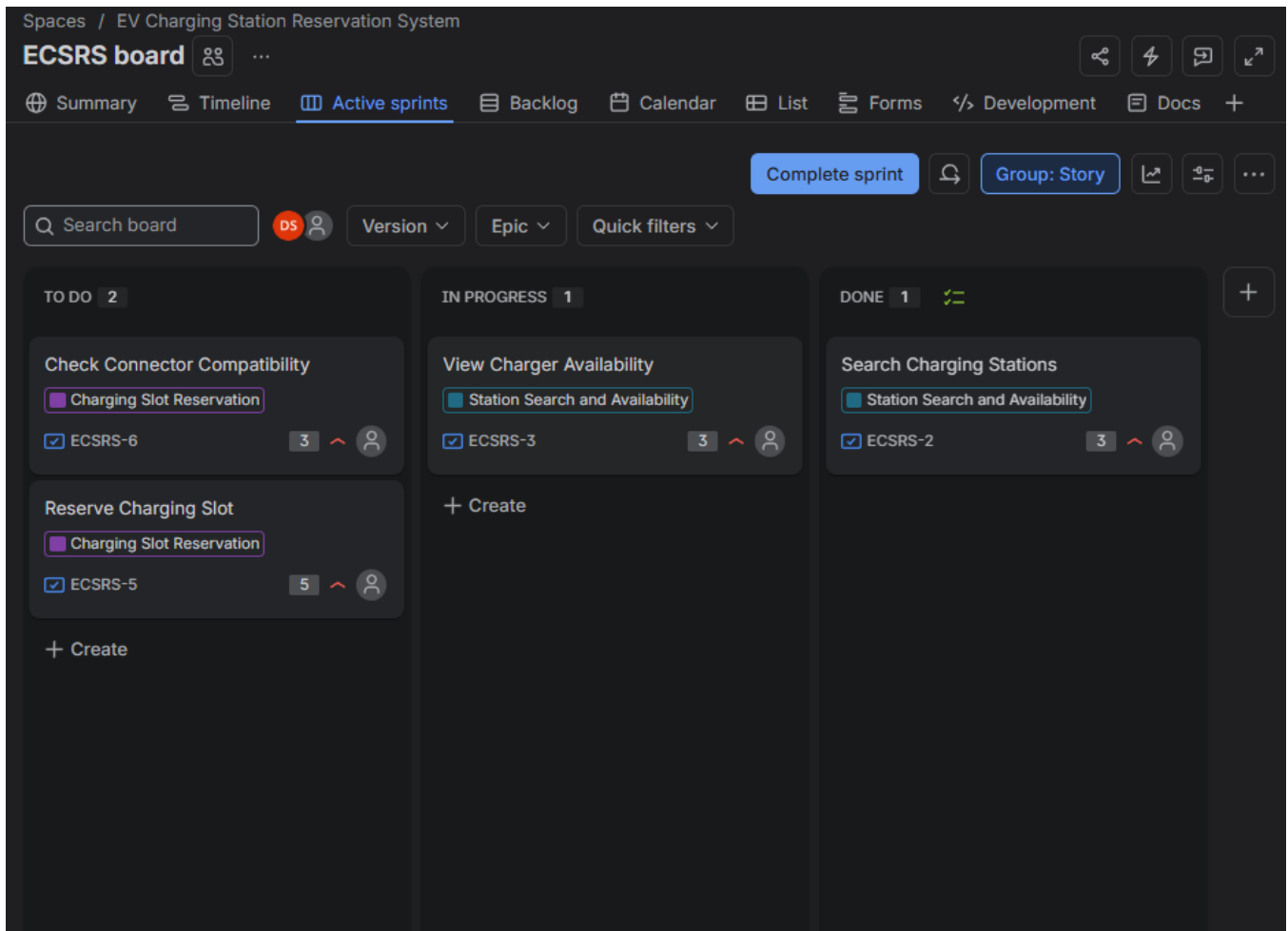


Figure 2. Sprint 1 Active Sprint view during the simulation.

4. Jira Sprint 1 – Burndown Chart

The Sprint 1 Burndown starts at 14 story points and reaches 0 after all four stories are completed.

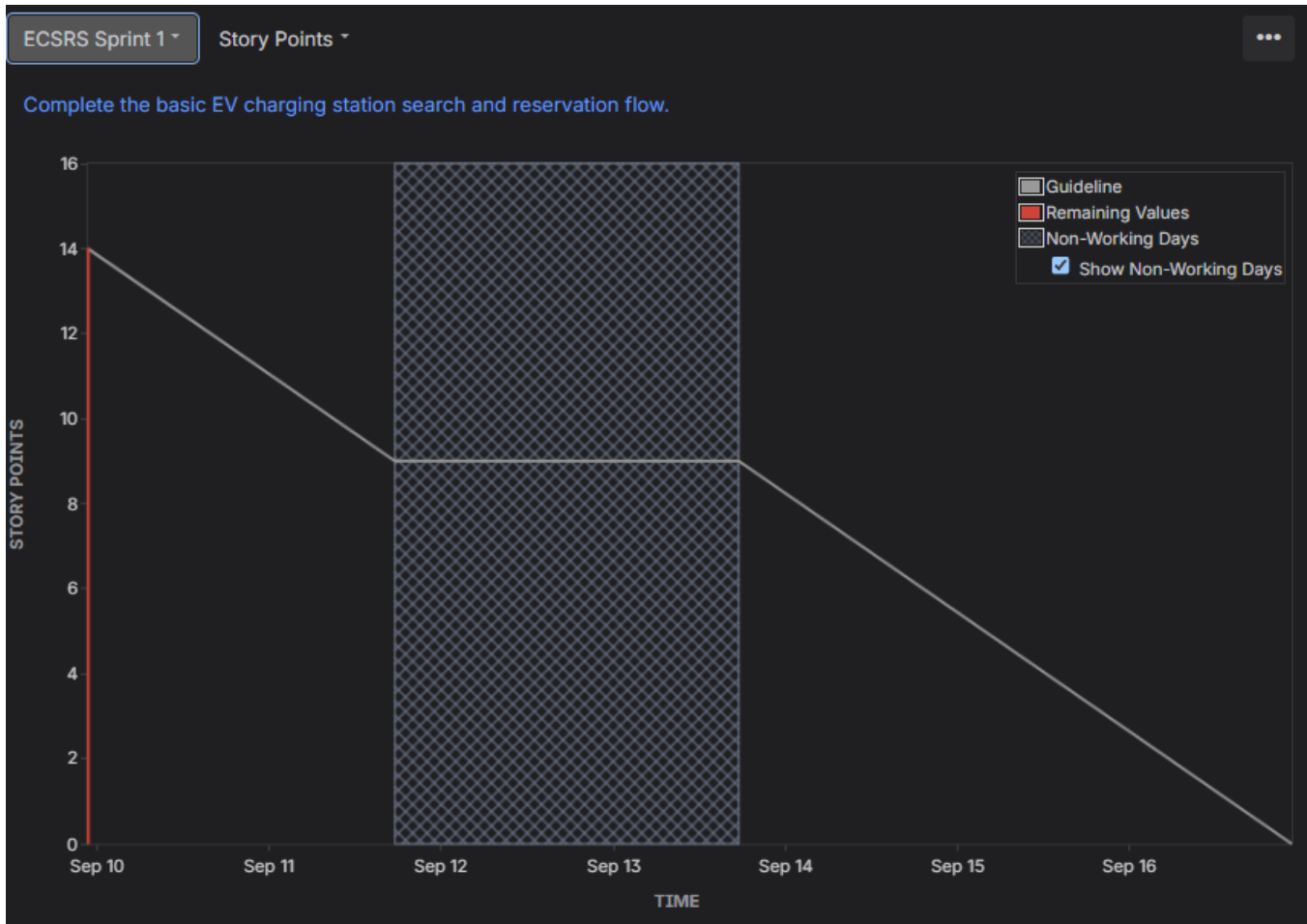


Figure 3. Sprint 1 Burndown Chart.

5. Jira Sprint 1 – Burndown Event Details

The event table records the individual completions and the corresponding remaining work: 14 → 11 → 8 → 5 → 0 story points.

Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
09/Sep/26 10:39 PM	ECSRS-2	Sprint start		3		
	ECSRS-3			3		
	ECSRS-5			5		
	ECSRS-6			3		
	Total					14
09/Sep/26 10:42 PM	ECSRS-2	Burndown	Issue completed		3	11
	ECSRS-3	Burndown	Issue completed		3	8
	ECSRS-6	Burndown	Issue completed		3	5
	ECSRS-5	Burndown	Issue completed		5	0
09/Sep/26 10:43 PM	ECSRS-2	Sprint ended by DHANUSH K S				0
	ECSRS-3					0
	ECSRS-5					0
	ECSRS-6					0
	Total					0

Figure 4. Sprint 1 Burndown event details.

6. Jira Sprint 2 – Active Sprint

Sprint 2 contained the four remaining stories totaling 16 story points. The Active Sprint view below shows the completed Sprint 2 work.

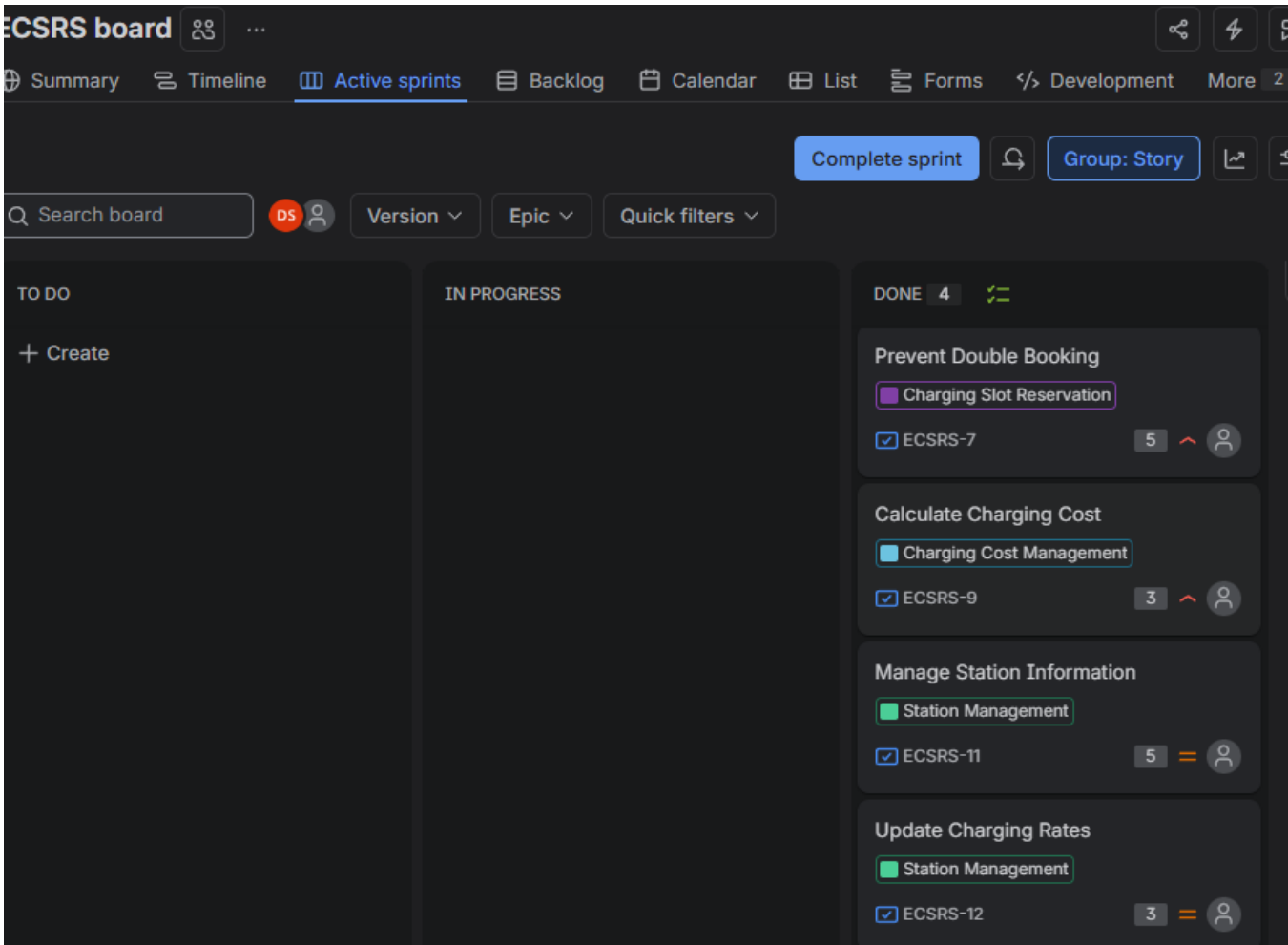


Figure 5. Sprint 2 Active Sprint view showing all four stories in Done.

7. Jira Sprint 2 – Burndown Chart

After correcting the Sprint 2 dates and re-running the completion sequence, the Burndown records 16 → 11 → 8 → 3 → 0 remaining story points.

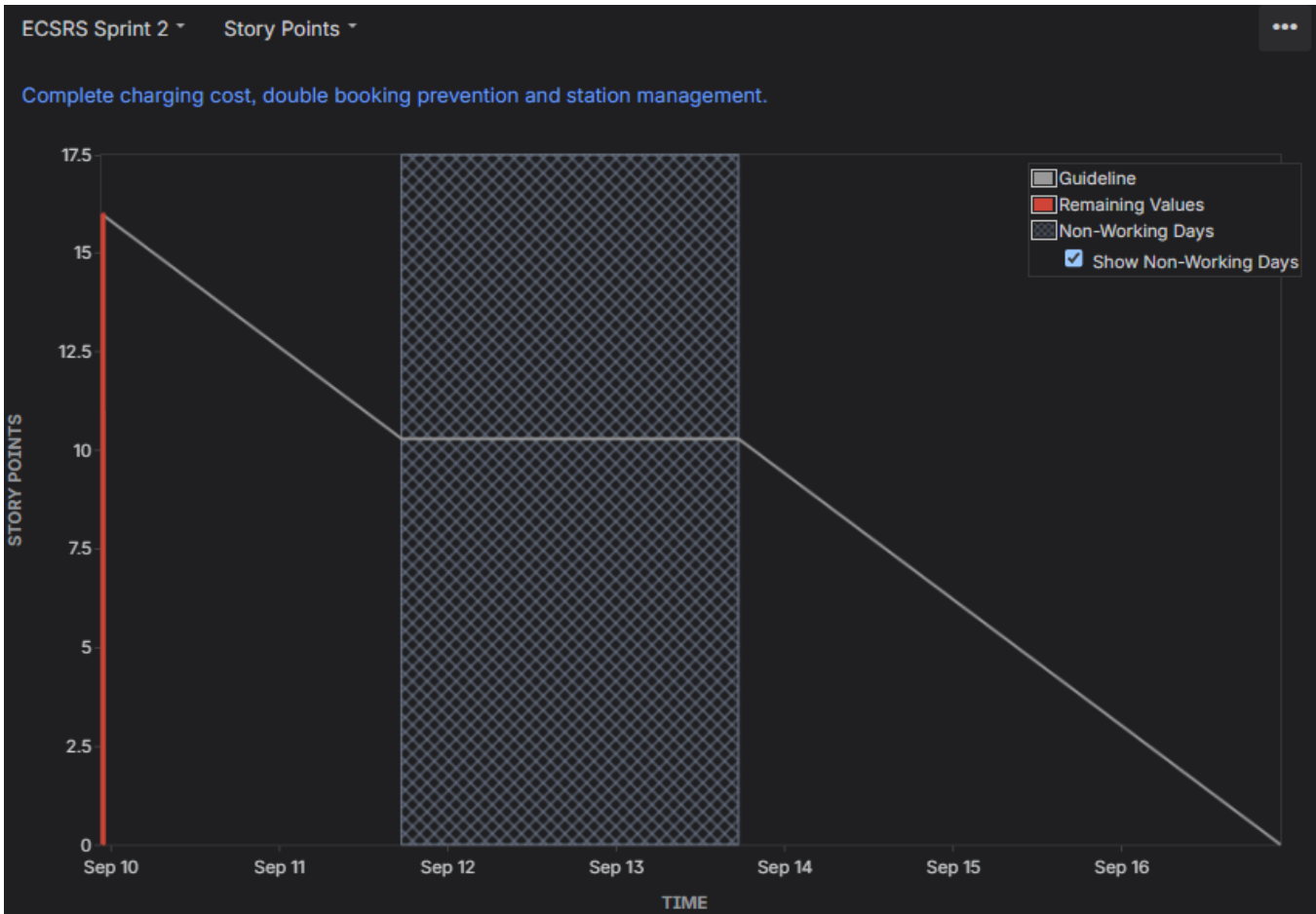


Figure 6. Corrected Sprint 2 Burndown Chart.

8. Jira Sprint 2 – Burndown Event Details

The Jira event table provides supporting evidence for the corrected Sprint 2 completion sequence. The remaining Story Points reduce from 16 to 11, then 8, then 3, and finally 0.

Date	Issue	Event Type	Event Detail	Story Points		
				Sts.	Qty.	Remaining
06/Sep/26 10:45 PM	ECSPRS-11	Sprint start		5		16
	ECSPRS-12			3		
	ECSPRS-7			5		
	ECSPRS-9			3		
	Total					
06/Sep/26 10:48 PM	ECSPRS-7	Burndown	Issue completed	5	11	
	ECSPRS-9	Burndown	Issue completed	3	8	
	ECSPRS-11	Burndown	Issue completed	5	3	
	ECSPRS-12	Burndown	Issue completed	3	0	
06/Sep/26 11:00 PM		Sprint ended by DHANUSH K S				
06/Sep/26 11:01 PM		Sprint reopened by DHANUSH K S				
06/Sep/26 11:02 PM	ECSPRS-7	Burndown	Issue reopened	5	5	
	ECSPRS-9	Burndown	Issue reopened	3	8	
	ECSPRS-11	Burndown	Issue reopened	5	13	
	ECSPRS-12	Burndown	Issue reopened	3	16	
06/Sep/26 11:05 PM	ECSPRS-7	Burndown	Issue completed	5	11	
06/Sep/26 11:06 PM	ECSPRS-9	Burndown	Issue completed	3	8	
06/Sep/26 11:07 PM	ECSPRS-11	Burndown	Issue completed	5	3	
	ECSPRS-12	Burndown	Issue completed	3	0	
	ECSPRS-11	Sprint ended by DHANUSH K S				
	ECSPRS-12					
	ECSPRS-7					
	ECSPRS-9					
	Total					0

The event table shows the corrected Sprint 2 completion records, including the final remaining value of 0.

Figure 7. Sprint 2 Burndown event details showing the corrected completion sequence.

9. Reflection Questions

1. Did your estimations reflect the actual effort?

The Fibonacci Story Points provided a useful relative estimate of complexity and uncertainty. Stories such as searching stations and checking connector compatibility were estimated at 3 points, while reservation, double-booking prevention and station information management were estimated at 5 points. Since this was a simulation, the points represent relative effort rather than actual development hours.

2. Was your backlog well-prioritized?

Yes. Core EV-driver functions such as searching, availability, compatibility, reservation, double-booking prevention and charging cost were given High priority. Station-operator maintenance functions were kept at Medium priority because they support correct station information and charging rates.

3. How did your simulated sprint align with your plan?

The backlog was divided into two focused one-week sprints. Sprint 1 completed search, availability, compatibility and reservation for 14 points. Sprint 2 completed double-booking prevention, charging cost and station-management work for 16 points. All eight backlog stories were completed.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Charts clearly showed the reduction of remaining work as stories were completed. Sprint 1 reduced 14 points to 0, while the corrected Sprint 2 sequence reduced 16 points to 0. This helps evaluate whether the amount of work committed to a sprint is realistic.

10. Conclusion

This lab demonstrated how Lab 1 functional requirements can be transformed into an Agile backlog using Epics and User Stories. Jira was used to prioritize work, assign Fibonacci Story Points, organize two one-week sprints, track progress through To Do → In Progress → Done, and analyse sprint progress using Burndown Charts.

Final evidence included: Jira backlog, Story Point assignments, one Active Sprint screenshot for Sprint 1, one Active Sprint screenshot for Sprint 2, both Burndown Charts, Burndown event details for both sprints, and all four reflection answers.